

Lab: The Paper Search Engine

Goal

To understand how a Search Engine Indexes information.

What You Need

- 20 Index Cards.
- Books/Magazines.

Steps

1. The Indexing (The Crawler)

Look through the books. Write down a keyword on a card ("Dinosaur", "Space", "Cake").
Write the Page Number where you found it.

2. The Sorting

Put the cards in Alphabetical Order. This is what Google does. It organizes the words so it can find them fast.

3. The Search

Ask a friend: "Find me a Dinosaur." Instead of looking through every book, look at the Index Cards. D -> Dinosaur -> Page 42. Boom! Fast retrieval.

Discussion

- How long would it take to find the dinosaur without the index?
- This is why we need computers to organize the world's data.

What We Learned

Algorithms help us organize chaos into order.